2.45

$\rho(r) = kr$

$$W = \frac{\varepsilon_0}{2} \int \vec{E}^2(r)\, d^3\vec{r}$$

Find $\vec{E}$: 1) $r > R$ $\quad E(r) \cdot 4\pi r^2 = \frac{Q_{TOTAL}}{\varepsilon_0}$

$$Q_{TOTAL} = \int_0^R (kr)\, 4\pi r^2 dr = 4\pi k \int_0^R r^3 dr =$$

$$= 4\pi k \frac{R^4}{4} = \pi k R^4.$$

$$\vec{E} = \frac{kR^4}{4\varepsilon_0 r^2}\,\hat{r}$$

2) $r < R \Rightarrow 4\pi r^2 E(r) = \frac{\pi k r^4}{\varepsilon_0}$ (charge inside sphere of radius r)

$$\vec{E} = \frac{kr^2}{4\varepsilon_0}\,\hat{r}$$

$$W = \frac{\varepsilon_0}{2} \int_0^R \left(\frac{kr^2}{4\varepsilon_0}\right)^2 4\pi r^2 dr$$

$$+ \frac{\varepsilon_0}{2} \int_R^\infty \left(\frac{kR^4}{4\varepsilon_0 r^2}\right)^2 4\pi r^2 dr =$$

$$= \frac{\varepsilon_0}{2} \left(\frac{k}{4\varepsilon_0}\right)^2 4\pi \left\{ \underbrace{\int_0^R r^6 dr}_{\frac{1}{7}R^7} + \int_R^\infty R^8 \frac{dr}{r^2} \right\} = \frac{\pi k^2}{7\varepsilon_0} R^7.$$

$$R^8 \cdot \frac{1}{R} = R^7$$

2.46 $V(r) = A\dfrac{e^{-\lambda r}}{r}$

$$\vec{E} = -\vec{\nabla} V = -\hat{r}\,\frac{\partial V}{\partial r} \quad \text{(since } V \text{ depends only on } r\text{)}$$

$$\vec{E} = -\hat{r}\,A\left(-\frac{e^{-\lambda r}}{r^2} - \frac{\lambda e^{-\lambda r}}{r}\right) = \hat{r}\,A\,\frac{e^{-\lambda r}}{r}\left(\lambda + \frac{1}{r}\right)$$

$$\rho = \varepsilon_0\,\vec{\nabla}\cdot\vec{E} = A\varepsilon_0\,\vec{\nabla}\cdot\left(\frac{\hat{r}}{r^2}\,e^{-\lambda r}(1+\lambda r)\right) =$$

$$= A\varepsilon_0\left\{\underbrace{\left(\vec{\nabla}\cdot\frac{\hat{r}}{r^2}\right)}_{4\pi\delta^3(r)}\left[e^{-\lambda r}(1+\lambda r)\right] + \frac{\hat{r}}{r^2}\cdot\left(\vec{\nabla}\,e^{-\lambda r}(1+\lambda r)\right)\right\}$$

$$= A\varepsilon_0\left\{4\pi\delta^3(r) + \frac{1}{r^2}\,\frac{\partial}{\partial r}\left(e^{-\lambda r}(1+\lambda r)\right)\right\}$$

$$\frac{1}{r^2}\left(-\lambda e^{-\lambda r}(1+\lambda r) + \lambda e^{-\lambda r}\right) = -\frac{\lambda^2}{r}e^{-\lambda r}$$

$$\rho = A\varepsilon_0\left(4\pi\delta(\vec{r}) - \frac{\lambda^2}{r}e^{-\lambda r}\right)$$

Total charge $Q = \int \rho\, d^3\vec{r} = A\varepsilon_0\left\{\int 4\pi\delta(\vec{r})\,d^3r\right.$

$$\left. - \lambda^2\int_0^\infty \frac{e^{-\lambda r}}{r}\,4\pi r^2 dr\right\} = A\varepsilon_0\left\{4\pi - 4\pi\lambda^2\int_0^\infty e^{-\lambda r}\,r\,dr\right\}$$

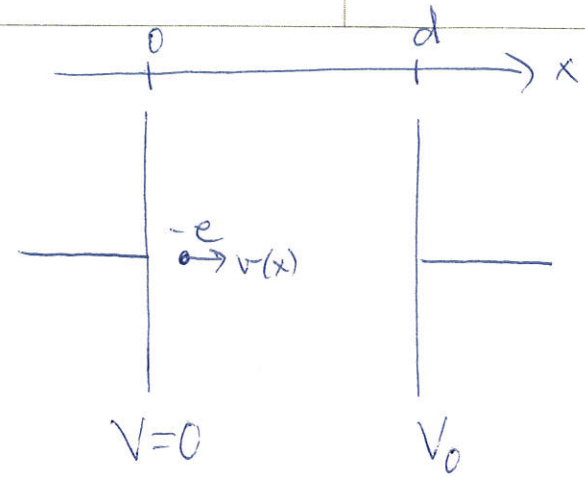
$$= A\varepsilon_0\left\{4\pi - 4\pi\underbrace{\int_0^\infty e^{-x}\,x\,dx}_{1}\right\} = 0$$

zero total charge!

$x = \lambda r$

Positive nucleus is screened by electrons cloud.

P2.48



a) $\nabla^2 V = -\frac{\rho}{\varepsilon_0} \Rightarrow \frac{d^2 V}{dx^2} = -\frac{\rho}{\varepsilon_0}$ (neglecting variation of V with y, z)

b) Energy $E = \frac{1}{2} m v^2 + (-e V) = \text{const}$

At $x=0$ electrons start from rest, and $V(x=0)=0$

$\Rightarrow \text{const} = 0 \Rightarrow e V(x) = \frac{1}{2} m v^2(x)$

↳ speed at x.

potential at x

c) Current $I(x) = \text{Area} \cdot \text{charge density} \cdot \text{speed}$

$= A(+\rho) v$ (Note that $\rho < 0$ and $I < 0$ because electrons are negatively charged.)

$\Rightarrow$ for steady current $I(x) = I = \text{const}$

$$\Rightarrow v(x) = +\frac{I}{A\rho}$$

$$\text{or } \rho(x) = +\frac{I}{A v(x)}$$

d) $$\frac{d^2 V}{dx^2} = \frac{-I}{\varepsilon_0 A v(x)} = -\frac{I}{\varepsilon_0 A} \frac{1}{\sqrt{\frac{2e}{m} V(x)}} = \frac{C}{\sqrt{V}}$$

where $C = \frac{-I}{\varepsilon_0 A}\sqrt{\frac{m}{2e}} > 0.$

Try $V(x) = \beta x^{\alpha}$ (since we know that $V(0)=0$)

Then $\dfrac{d^2 V}{dx^2} = \beta\, \alpha(\alpha-1)\, x^{\alpha-2}$

$$\frac{1}{\sqrt{V}} = \frac{1}{\sqrt{\beta}}\, x^{-\alpha/2}$$

$$\Rightarrow \quad \beta\, \alpha(\alpha-1)\, x^{\alpha-2} = \frac{C}{\sqrt{\beta}}\, x^{-\alpha/2}$$

$$\beta^{3/2}\, \alpha(\alpha-1)\, x^{\alpha-2} = C\, x^{-\alpha/2}$$

$\Rightarrow$ need $x^{\alpha-2} = x^{-\alpha/2} \Rightarrow \alpha - 2 = -\frac{\alpha}{2} \Rightarrow \frac{3}{2}\alpha = 2.$

$$\alpha = \frac{4}{3}.$$

and $C = \beta^{3/2}\, \frac{4}{3}\left(\frac{4}{3}-1\right) = \frac{4}{9}\beta^{3/2}$, $\beta = \left(\frac{9C}{4}\right)^{2/3}$

Since $C = -\dfrac{I}{\varepsilon_0 A}\sqrt{\dfrac{m}{2e}}$, $\beta = \left(-\dfrac{9I}{4\varepsilon_0 A}\right)^{2/3}\left(\dfrac{m}{2e}\right)^{1/3}$

Thus: $\left.\begin{aligned} V(x) &= \beta\, x^{4/3} \\ V(d) &= \beta\, d^{4/3} = V_0 \end{aligned}\right\}$ $V(x) = V_0\left(\dfrac{x}{d}\right)^{4/3}$,

non-linear $V(x)$!

$$\Rightarrow \quad \left(\frac{-9I}{4\varepsilon_0 A}\right)^{2/3} (d^2)^{2/3}\left(\sqrt{\frac{m}{2e}}\right)^{2/3} = V_0$$

$$\frac{-9I\, d^2}{4\varepsilon_0 A}\sqrt{\frac{m}{2e}} = V_0^{3/2}$$

$$-I = |I| = \underbrace{\frac{4\varepsilon_0 A}{9d^2}\sqrt{\frac{2e}{m}}}_{K}\, V_0^{3/2}$$

Nonlinear I-V characteristics.

Note that in the <u>absence of space-charge</u> ($\rho = 0$) the Poisson's eq. transforms into Laplace's one:

$$\frac{d^2 V}{dx^2} = 0$$

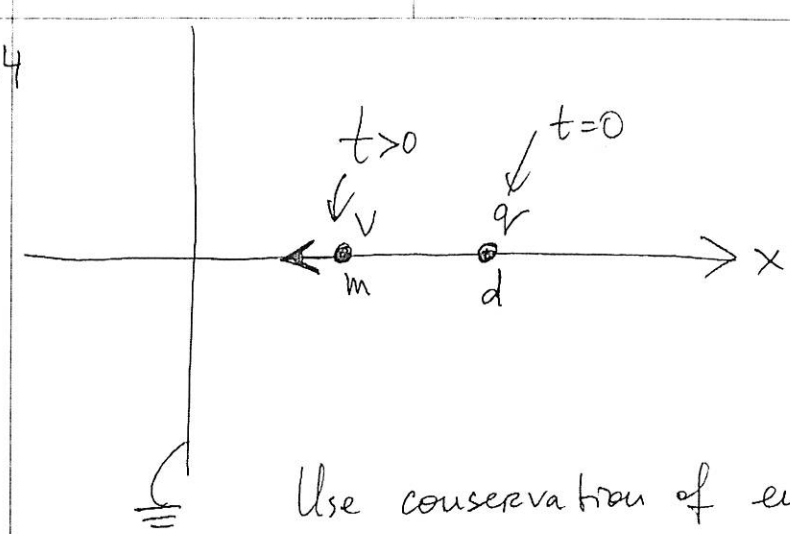
General solution: $V(x) = a + bx$

BC: $V(0) = 0 \Rightarrow a = 0$

$V(d) = V_0 \Rightarrow bd = V_0 \Rightarrow b = \frac{V_0}{d}$

$\Rightarrow V(x) = V_0 \left(\frac{x}{d}\right)$ ← linear dependence of V on x

P3.34



Use conservation of energy:

$$E_{initial} = -\frac{1}{4\pi\varepsilon_0}\frac{q^2}{4d}$$

$$E(t) = -\frac{1}{4\pi\varepsilon_0}\frac{q^2}{4x} + \frac{1}{2}mv^2 \quad , \text{for } x<d.$$

$$\vec{v} = \frac{dx}{dt}\hat{x}$$

$$\Rightarrow \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 = \frac{q^2}{16\pi\varepsilon_0}\left(\frac{1}{x}-\frac{1}{d}\right)$$

choose negative sign:

$$\frac{dx}{dt} = -\sqrt{\frac{q^2}{8\pi\varepsilon_0 m}}\left(\frac{1}{x}-\frac{1}{d}\right)^{1/2}$$

$$\int_d^0 \frac{dx}{\sqrt{\frac{1}{x}-\frac{1}{d}}} = -\frac{q}{\sqrt{8\pi\varepsilon_0 m}}\int_0^t dt = -\frac{qt}{\sqrt{8\pi\varepsilon_0 m}}$$

$x = ds$

$$\int_1^0 \frac{d\,ds}{\frac{1}{\sqrt{d}}\sqrt{\frac{1}{s}-1}} = d^{3/2}\int_1^0 \frac{\sqrt{s}\,ds}{\sqrt{1-s}}$$

let now $s = \cos^2\alpha \Rightarrow I = +\int_0^{\pi/2} \frac{\cos\alpha\, d(\cos^2\alpha)}{\sqrt{1-\cos^2\alpha}} =$

$$= -\int_0^{\pi/2} \frac{\cos\alpha}{\sin\alpha}\, 2\cos\alpha\sin\alpha\, d\alpha = -2\int_0^{\pi/2}\cos^2\alpha\, d\alpha = -2\left(\frac{\pi}{4}\right) = -\frac{\pi}{2}$$

Thus $\left(-\frac{\pi}{2}\right) d^{3/2} = -\frac{q\,t}{\sqrt{8\pi\varepsilon_0 m}}$

$$t = \frac{\pi}{2}\frac{\sqrt{8\pi\varepsilon_0 m}}{q}\, d^{3/2} = \frac{\pi d^{3/2}}{q}\sqrt{2\pi\varepsilon_0 m}$$